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Infrared thermography as an adjunctive tool for detection of femoral arterial thrombosis after cardiac catheterization: A prospective, pilot study

Sok-Leng Kang MBBS, MSc | Larissa Manojlovich MD | Dariusz Mrozcek MD | Lee Benson MD

Department of Pediatrics, Division of Cardiology, The Labatt Family Heart Centre, The Hospital for Sick Children, Temerty School of Medicine, Toronto, Ontario, Canada

Correspondence

Lee Benson, MD, Department of Pediatrics, Division of Cardiology, The Labatt Family Heart Center, The Hospital for Sick Children, 555 University Ave, Toronto, ON M5G 1X8, Canada.
Email: lee.benson@sickkids.ca

Present address

Sok-Leng Kang, MBBS, MSc, Department of Pediatric Cardiology, Alder Hey Children's Hospital, Liverpool, UK.

Abstract

Objectives: To assess the utility of infrared thermography (IRT), to map skin temperature, in the detection of femoral arterial (FA) thrombosis after cardiac catheterization.

Background: Ultrasound is a validated method for thrombus detection but is generally reserved as a confirmatory test for clinical suspicion due to various constraints.

Methods: Prospective study of infants and children undergoing cardiac catheterization via FA access, comparing IRT and pulse examination. The thermograms, displayed in a color map with each pixel representing a temperature, were examined by qualitative assessment of symmetry in thermal patterns and quantitative image analysis with abnormal thermographic asymmetry defined as a difference of >10% between limbs.

Results: In the 20 children enrolled, excellent agreement was found between the two methods with a Kappa value of 0.89. The median thermographic asymmetry in the nine children with pulse loss was 36 (13–76)%. Using receiver operating characteristic analysis, the asymmetrical pattern of ≥18% between limbs predicted the need for anticoagulation with a sensitivity of 100% and specificity of 89%. The area under the curve was 0.97 (95% confidence interval: 0.95–1). Children with absent pulse requiring anticoagulation showed a slower recovery in thermal asymmetry compared to those with a reduced pulse. By qualitative IRT assessment, all children with absent pulse requiring anticoagulation were correctly identified by 10 independent assessors.

Conclusions: This pilot study showed that IRT is feasible and reliable as an adjunctive tool for thrombus detection postcatheterization and treatment monitoring. Specific advantages of IRT include portability, affordability, and contactless image acquisition.

KEYWORDS

cardiac catheterization, children, infants, thermography, thrombosis

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1 | INTRODUCTION

Early detection and prompt treatment of femoral arterial thrombosis after cardiac catheterization is critical to prevent acute limb ischemia and the longer-term adverse consequences on leg development. In current practice, the clinical assessment remains the mainstay of diagnosis and the reported incidence of arterial thrombosis postcardiac catheterization ranges from 0.6% to 9.6%.[1–12] While ultrasound screening has been shown to increase thrombus detection rates particularly in infants[13,14]; it is generally reserved as a confirmatory test due to time and resource constraints. To this end, novel application of infrared thermography (IRT), a noncontact, affordable, and portable imaging technique to measure heat radiation may provide an adjunctive diagnostic tool.[15] Because vascular integrity is a major determinant of skin temperature, infrared cameras can be used to detect thermal variation in the limbs in the event of femoral arterial occlusion.[15,16] In this prospective pilot study, we assessed the utility of IRT in the detection of occlusive femoral arterial injury in children after cardiac catheterization and compared it to clinical examination. We hypothesized that IRT would have good agreement with the physical examination and may predict the need for anticoagulation in children with suspected femoral arterial thrombosis postcardiac catheterization.

2 | METHODS

2.1 | Study design

This prospective study was approved by the institutional ethics review board. Children aged 0–16 years who require cardiac catheterization through femoral arterial access over a 9-month period were approached to participate in the study and informed consent was obtained. Patients with previously identified arterial or venous thrombosis of the lower limbs, infection or inflammation of the lower limbs, known rheumatic or vascular disorders such as juvenile idiopathic arthritis and vasculitis, were excluded. All children underwent clinical assessment and IRT before and within 30–60 min after the catheterization. If the pulse check was normal during recovery, a final IRT was performed at 4–6 h after the catheterization before discharge. If a reduced/absent pulse was detected at the initial postcatheterization assessment, repeat IRT was performed at intervals of 1–2 hourly up to 6 h after the study. The clinical team was blinded to the findings of the IRT, and all treatment decisions were based solely on clinical assessment guided by the institutional protocol for "Management of femoral artery thrombosis." (Appendix SI) In brief, initiation of heparin therapy was based on *clinical diagnosis* of an absent pulse. Vascular ultrasound was performed acutely if there were signs of limb ischemia or in the absence of recovery of pulses the day following the procedure, in preparation for longer-term anticoagulation.

2.2 | Clinical assessment

Pedal pulse assessment was categorized as normal or abnormal. *A normal* examination was defined as equal pulses in both limbs. An *abnormal* examination included *reduced* and *absent pulse* compared to the uncannulated limb. *Limb ischemia* was defined as a significantly cool, pale extremity with a delayed or absent capillary refill.

2.3 | Infrared thermography

Thermograms of both lower limbs were obtained by a single investigator (S. L. K.) using a portable infrared camera (Seek Thermal) attachable to an Android phone (Google). The Seek compact Pro provides a pixel resolution of 320 × 240 and thermal sensitivity of <70 mK (i.e., <0.07°C sensitivity). The lower limbs were exposed to room temperature and potential sources of external warming were removed before IRT. The thermal camera was kept perpendicular to the plane of acquisition and the imaging distance was kept <1 m. Additional camera specifications and technical considerations are provided in Appendix SII. Thermograms were displayed in "pseudocolor" with each individual pixel representing a temperature. A color–temperature scale bar is displayed for each thermogram and the temperature ranges depend on auto-calibration against the background ambient temperature. A color scheme "spectra" was used in this study, with blue and green representing lower temperatures, and the brighter colors of yellow, orange, red, and white representing higher temperatures (Figure 1)

Analysis of the thermograms consisted of:

1. *Qualitative analysis*: The cannulated and uncannulated limbs were simply compared to determine general symmetry (Figure 1), as published normative data in adults has shown symmetrical thermographic patterns between the same regions in contralateral limbs.[16]
2. *Quantitative analysis*: Temperature data in the thermogram could not be extracted using the Seek Thermal software. Thus, to quantify the area of thermal asymmetry between the lower limbs, the thermograms were exported as JPEG images to ImageJ software. The methodology for image analysis is provided in Appendix SIII. In brief, areas representing cooler temperatures in each lower limb were segmented using the "color thresholding" technique and expressed as a percentage of the total surface area of the limb (Figure 2). *Abnormal thermogram* is defined as a difference of >10% between limbs. Based on bench testing, this cut-off was decided a priori to account for minor differences between limbs that could be attributed to environmental factors or measurement error.

The thermograms were analyzed by a single investigator (L. M.) who was blinded to the results of clinical assessment. All measurements were performed twice, 1 month apart and the average value was used for analysis. To assess interobserver variability another investigator (S. L. K.) performed measurements on all initial postcatheterization thermograms in random order. To determine if

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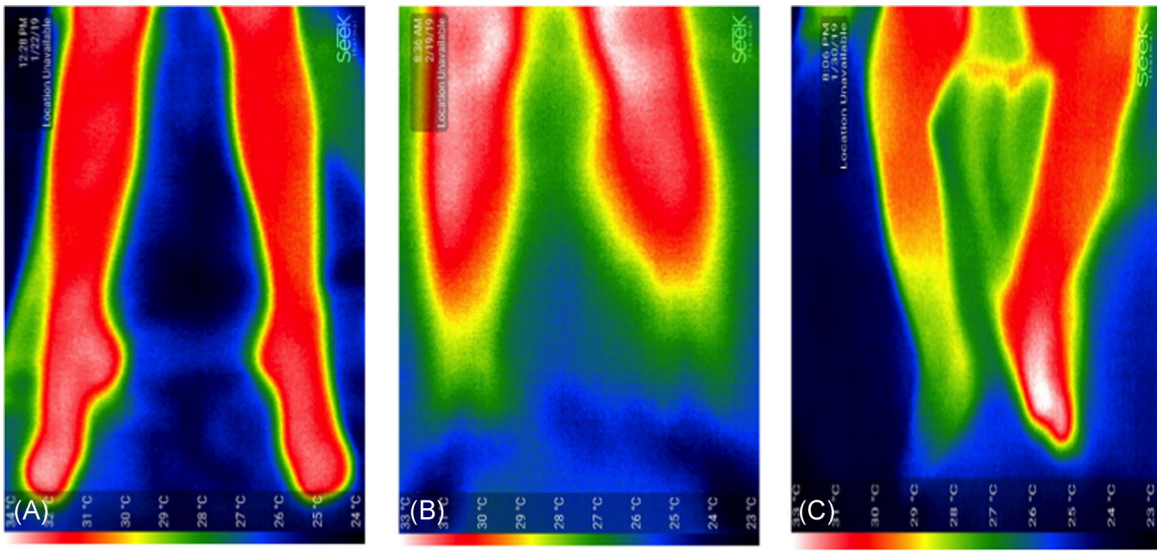


FIGURE 1 Examples of thermograms displayed using the color scheme "spectra." The color–temperature scale bar is displayed for each image, with blue and green representing lower temperatures, and brighter colors of yellow, orange, red, and white representing higher temperatures. Qualitative image analysis: (A) Thermogram of a patient obtained precatheterization showing a symmetrical thermal pattern. (B) Thermogram of a patient with aortic coarctation preintervention showing cool feet but a symmetrical thermal pattern. (C) Thermogram of a patient post right femoral arterial cannulation for patent ductus arteriosus stenting, showing asymmetric thermal distribution of lower limbs [Color figure can be viewed at wileyonlinelibrary.com]

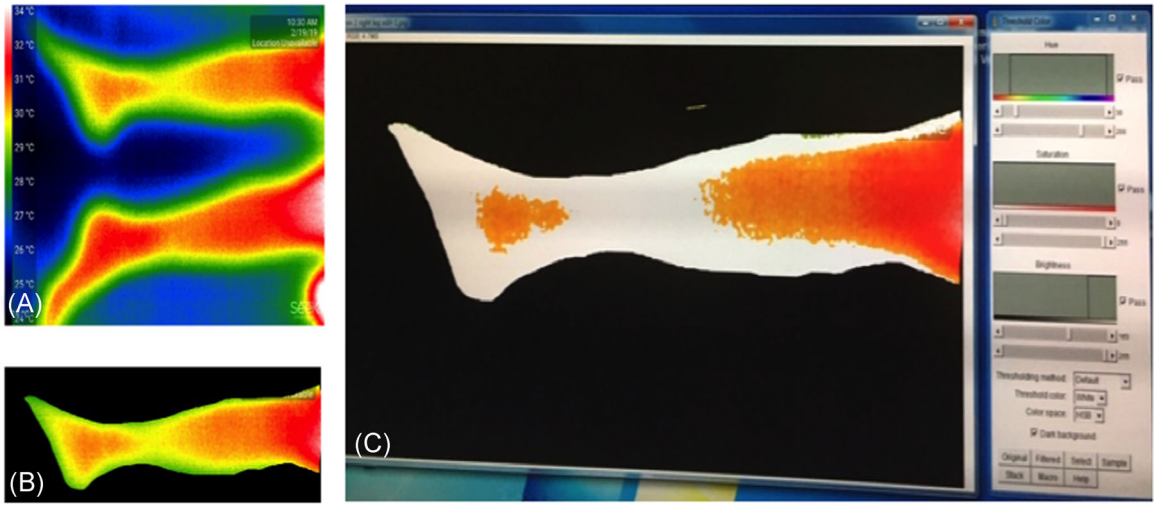


FIGURE 2 Quantitative image analysis: (A) The thermal image was exported to ImageJ as a JPEG file. (B) The right leg is segmented and isolated. The total area of the leg is measured by an automatic calculation of the number of pixels within the area of interest. (C) Using the color thresholding function, colors corresponding to cool temperatures were automatically segmented (white areas) and these areas were again measured. The proportion of cool areas to that of the whole limb was then calculated. The process is repeated with the left leg. Thermal asymmetry is expressed as the difference in proportions of cool areas between the two limbs [Color figure can be viewed at wileyonlinelibrary.com]

clinically significant thermographic abnormalities could be detected by qualitative assessment alone, nurses in the cardiac catheterization laboratory and recovery unit were asked to review three thermograms (before and after catheterization and before discharge) for each child in sequence and identify those consistent with an abnormal pulse examination (absent/reduced pulse). The arterial access site was indicated on each thermogram.

2.4 | Statistical analysis

The outcome was to compare the result of the IRT versus clinical examination. Agreement between IRT and clinical examination was assessed with Cohen's Kappa coefficient. A receiver operating characteristic (ROC) curve analysis was performed to determine the discriminatory ability of IRT and cut-off for prediction of the need for

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anticoagulation (or finding of an absent pulse).[17] Intra- and interobserver agreement for the quantitative approach was assessed by an intraclass correlation coefficient (ICC). Multirater agreement for the qualitative approach was examined with a Fleiss Kappa. Statistical calculations were performed with SPSS version 26 (IBM)

3 | RESULTS

Over the 9-month period, twenty children were enrolled in the study. Demographic characteristics, procedural details, access vessel complications, and outcomes are summarized in Table 1. At the initial postcatheterization examination, five children had absent pulses and were treated with a heparin infusion. Full restoration of pulse was observed in two and persistent pulse loss in three children. Subsequent vascular ultrasound confirmed an occlusive ($n = 2$) and nonocclusive ($n = 1$) thrombus (Table 1). Four other children had reduced pulses, which recovered at the subsequent examinations within 4 h postcardiac catheterizations.

A total of 68 thermograms were obtained as outlined in the study design, 17 of which were considered abnormal based on the predefined criteria of thermal asymmetry. The median thermographic asymmetry in the nine children with pulse loss was 36 (13–76)%. The associations between thermographic findings and pulse examination are depicted in Table 2. There was excellent agreement between the two methods with a κ value of 0.89.

ICC was applied to assess intra- and interobserver variation in the quantitative approach and showed excellent agreement between measurements performed by a single assessor ($n = 68$ pairs, ICC = 0.997, 95% confidence interval [CI]: 0.995–0.998) and between two assessors ($n = 20$ pairs, ICC = 0.993, 95% CI: 0.981–0.997)

Further analysis of the thermograms was performed to determine a cut-off point of thermal asymmetry that would reliably identify all children with absent pulse requiring anticoagulation. Based on the ROC curve analysis, a difference of ≥18% in the thermal pattern was identified as the optimal threshold to detect a clinically absent pulse with a sensitivity of 100% and specificity of 89%. The area under the curve was 0.97 (95% CI: 0.95–1), indicating excellent diagnostic accuracy (Figure 3).

Further, we examined the changes in thermal pattern in consecutive thermograms acquired in eight of nine children with an abnormal pulse check at the initial postcatheterization assessment. One child did not have serial thermograms because the predefined IRT window was missed. The rate of change per hour in thermal asymmetry was calculated between the first and second thermograms postcatheterization. In the four children with a reduced pulse and no heparin treatment, there was a larger change toward baseline thermal pattern, with a median 80%, (range: 50%–90%) decrease in thermal asymmetry per hour from the initial abnormal thermogram to the next. In the other four children with absent pulse and on heparin therapy, there was either a worsening of thermographic asymmetry, no change or <50% decrease per hour.

Finally, 10 nurses assessed thermograms qualitatively for each child before and after the catheterization and identified abnormal thermograms that would indicate clinical pulse loss (Figure 4). There was a moderate overall agreement between the 10 raters with a Fleiss Kappa of 0.58 (95% CI: 0.577–0.582). However, all 10 assessors correctly identified the 5 children with an absent pulse requiring anticoagulation. In the remaining 15 children with a reduced or normal pulse after catheterization, the multirater agreement was fair, Fleiss Kappa of 0.24 (95% CI: 0.249–0.258).

4 | DISCUSSION

The first medical application of IRT was for preclinical diagnosis of breast cancer in 1956.[18] Since then, evolving technology has led to a steady increase in its application for diagnosis and treatment monitoring of various diseases including vascular disorder, liver metastatic disease, bowel ischemia, cerebrovascular, rheumatic, dermatological, and gynecological diseases.[15,19–22] Thermography is uniquely effective in the diagnosis of vascular disorders because skin temperature is closely related to the blood supply to cutaneous and subcutaneous tissue in addition to the integrity of vascular function. Furthermore, the human skin with an emissivity value of 0.98, is a near-ideal radiator in the infrared domain.[16] In this regard, there is emerging literature on the utility of IRT for detection of peripheral arterial disease (PAD), deep vein thrombosis (DVT), diabetic vasculopathy and neuropathy, posttraumatic acute compartment syndrome, and other vascular diseases.[20,23–33] Multiple studies have also demonstrated a good correlation between IRT and other established diagnostic tools, such as ankle-brachial index for diagnosis of PAD and Doppler ultrasound for DVT.

In this pilot study, we report a novel application of thermography for the detection of femoral arterial thrombosis in children after cardiac catheterization. Our study showed excellent agreement between IRT and the clinical palpation of pulses in determining the likelihood of thrombosis. Blinded to the outcome of the pulse check, assessment of the thermograms alone correctly identified all children who required anticoagulation following the detection of absent pulse. This was the case with both quantitative (estimation of thermal asymmetry on ImageJ) and qualitative (visual assessment by multiple assessors) approaches. There was excellent intra- and interobserver agreement using the quantitative method. Further ROC curve analysis showed that thermal asymmetry ≥18%, reliably predicted the need for anticoagulation with 100% sensitivity and 89% specificity. In practical terms, it can be inferred that if more than one-fifth of the affected limb appears significantly cooler than the contralateral healthy limb, there is a high likelihood of arterial thrombosis. Further, IRT can be useful to monitor disease progression or response to treatment with potential impact on clinical management. Analysis of consecutive thermograms from children with an abnormal pulse within the 6 h study period showed less improvement in thermal asymmetry per hour or worsening thermal asymmetry in children with thrombosis and on heparin therapy compared to those with the

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TABLE 1 Patient characteristics, procedure type and duration, and information related to diagnosis and treatment of femoral arterial thrombosis

Patient no.	Age (years)	Weight (kg)	Procedure	Arterial access site	Sheath size	Procedural duration (min)	Pulse assessment postprocedure	Heparin initiated (yes/no)	Outcome
1	0.07	4.1	Angiography of APC	LFA	4	131	Absent	Yes	Recovery of pulse
2	14.03	66.8	Aortic valve dilation	RFA	8	50	Normal	No	Recovery of pulse
3	9.87	32.5	CoA stent	RFA	12	64	Reduced	No	Recovery of pulse
4	7.30	25.8	CoA stent	RFA	12	79	Normal	No	Recovery of pulse
5	1.92	8.9	Attempted VSD closure	RFA	4	131	Absent	Yes	[a] Thrombus/LMWH
6	0.49	6.2	PDA stent	RFA	4	120	Absent	Yes	[a] Thrombus/LMWH
7	2.34	11.9	PDA closure	RFA	4	49	Normal	No	Recovery of pulse
8	6.90	18.1	CoA dilation	RFA	6	74	Reduced	No	Recovery of pulse
9	8.77	26.6	PDA closure	RFA	5	26	Normal	No	Recovery of pulse
10	0.33	5.1	BTS stent	RFA	4	44	Normal	No	Recovery of pulse
11	6.34	22	PDA	RFA	5	36	Normal	No	Recovery of pulse
12	17.06	42.7	PDA	RFA	5	54	Normal	No	Recovery of pulse
13	0.54	8.3	CoA dilation	RFA	4	32	Normal	No	Recovery of pulse
14	0.76	7.4	PDA closure	RFA	4	63	Absent	Yes	[a] Thrombus/LMWH
15	0.29	3.7	Aortic valve dilation	RFA	4	39	Reduced	No	Recovery of pulse
16	1.30	8.4	PDA closure	RFA	4	67	Reduced	No	Recovery of pulse
17	1.34	8.3	PDA closure	RFA	4	75	Absent	Yes	Recovery of pulse
18	16.28	96.4	Coronary angiography	RFA	6	46	Normal	No	Recovery of pulse
19	9.00	36	CoA stent	RFA	10	72	Normal	No	Recovery of pulse
20	10	26.5	Coronary angiography	RFA	6	45	Normal	No	Recovery of pulse

Abbreviations: APC, aortopulmonary collateral; BTS, Blalock–Tausig shunt; CoA coarctation of aorta; LFA, left femoral artery; PDA, patent ductus arteriosus; RFA, right femoral artery.

[a]Thrombus confirmed on ultrasound, treated with low molecular weight heparin.

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TABLE 2 Comparison of findings from thermography versus palpation of pulses

Infrared thermography	Palpation of pulse Thrombosis likely	Thrombosis unlikely
Thrombosis likely	17	2
Thrombosis unlikely	1	48
Total	18	50

$\kappa = 0.889$. The assessor of thermogram was blinded to results of pulse check. Thrombosis likely if thermographic asymmetry of >10% or clinical examination revealed absent or reduced pulse.

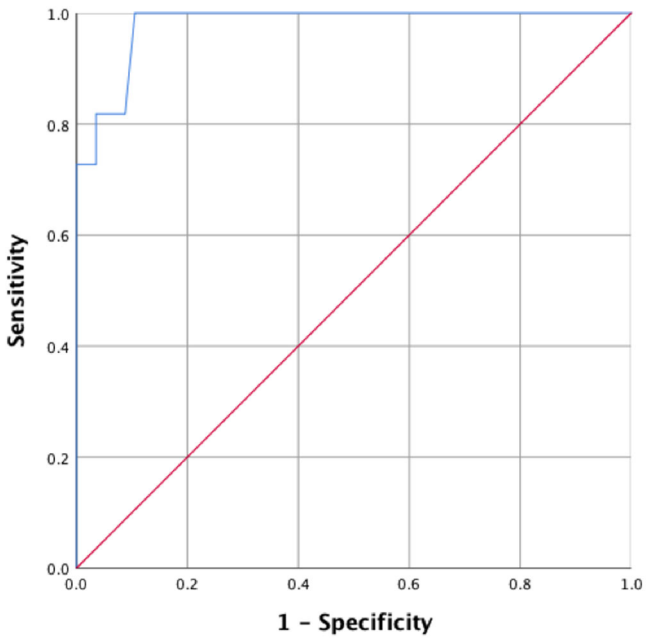


FIGURE 3 The ROC curve analysis of thermographic findings ($n = 68$) using nonparametric estimation method showed good diagnostic accuracy for the prediction of the need for anticoagulation. The area under the curve (AUC) was 0.97 (95% CI: 0.95–1). A difference of ≥18% in the thermal pattern was identified as the optimal threshold to detect an absent pulse with a sensitivity of 100% and specificity of 89%. CI, confidence interval; ROC, receiver operating characteristic [Color figure can be viewed at wileyonlinelibrary.com]

spontaneous recovery of pulses. In children who are discharged with low molecular weight heparin, serial thermograms performed at follow-up may provide useful information to assess response to therapy or adequacy of collateralization.

There are distinct advantages of IRT as an adjunctive diagnostic tool in the postcardiac catheterization environment. First, the technique is contactless and can be applied at the bedside, avoiding additional distress or discomfort to the infant or child. Second, image acquisition is simple and rapid, providing real-time information to indirectly assess recovery of pulses (improvement in thermal symmetry) or to monitor treatment response if anticoagulation or thrombolytic therapy was initiated. In instances where there is uncertainty in pulse check due to patient factors or clinicians' inexperience, a visual display of skin temperature IRT may provide additional information to support management decisions. Thirdly, the cost of a thermal camera is relatively inexpensive compared to other diagnostic modalities and a trained personnel is generally not required. However, the qualitative nature of image interpretation may raise concerns regarding the reliability of IRT. In our preliminary experience, clinically important thermographic asymmetry was readily discernable and in practice, IRT serves as a complementary modality, rather than a stand-alone assessment tool. As an imaging technique, IRT is also subjected to measurement error, which may be caused by camera performance, variation in field size, and angle of view relative to the position of the object of interest and environmental factors.[34] Thus, basic knowledge of the sources of error and attention to the image acquisition method is necessary and the use of high thresholds for clinically meaningful temperature differences are recommended. Recently, technological innovations such as portable smartphone-based devices have increased not only the accessibility and utility of IRT but importantly its accuracy and reliability when applied to the human body.[15,22] Further developments, such as establishing a common database of normal thermal images of all body parts, a shift to computer-aided diagnosis based on thermographic patterns and fusion of two-dimensional (2D) infrared images (which lack anatomical information) with 3D magnetic resonance imaging for superior diagnosis is expected to improve the clinical applicability and validity of IRT across many medical specialties.[15,35]

Limitations of this study include the nonavailability of ultrasound, considered the "gold standard" test for comparison in all children with pulse loss. Ultrasound was only obtained if there were signs of limb ischemia or in the absence of recovery of pulses despite heparin therapy, before discharge. The true incidence of thrombosis in our cohort is unknown because of the potential missed diagnosis of subclinical thrombosis, masked by sufficient collateralization around the occluded vessel. Additionally, various factors may affect the reliability of pulse examination, including operator's experience, patient cooperativeness, increased subcutaneous tissue, or presence of edema. While there were only a few discrepancies in the findings between IRT and pulse palpation in our series, the subject numbers were small. Additionally, temperature data in the thermogram could not be extracted using the Seek Thermal software. The JPEG images transferred to ImageJ contain only a visual representation of the temperature instead of the raw values.

Specific to our study purpose, further work will be necessary to gather more data on the thermographic distribution and patterns of recovery in the lower limb after femoral arterial cannulation in children with and without thrombus formation. Whilst ultrasound was only obtained in specific scenarios in the pilot study; future study incorporating ultrasound assessment in all children with pulse loss will enable correlation of IRT to ultrasound findings such as total or partial thrombosis, vessel spasm, or collateralization. In children with arterial thrombosis treated with longer-term anticoagulation, serial IRT at set intervals will allow monitoring of treatment response during follow-up. With the availability of large datasets from IRT

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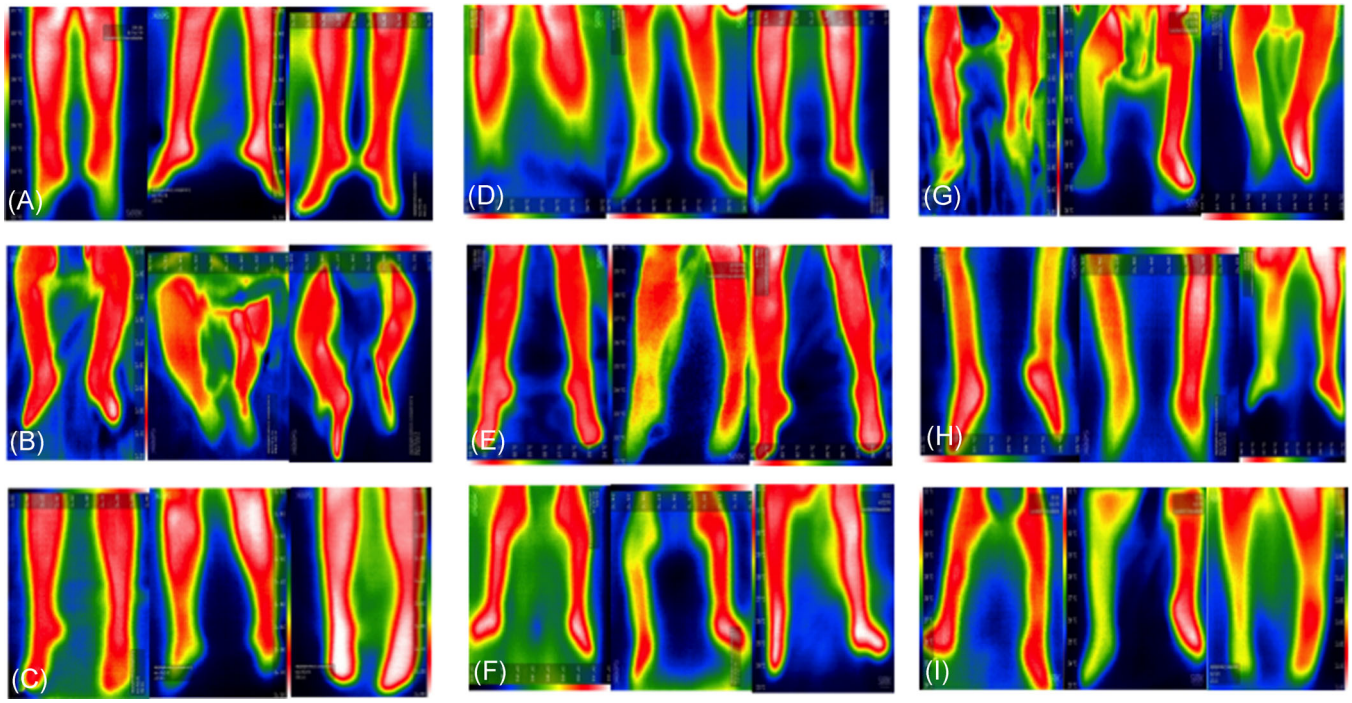


FIGURE 4 Examples of consecutive thermographic patterns for nine children. In each panel, the thermograms were acquired before the catheterization (left image); within 30–60 min (middle) and at 2–4 h after the catheterization (right). In Panels (A–C), there was no concern for pulse loss postcatheterization after a coarctation stent (A), Blalock–Taussig–Thomas shunt stent (B) and patent arterial duct (PDA) closure (C). In Panels (D–F), patients had reduced pulse after a coarctation stent (D), coarctation angioplasty (E), and PDA closure (F) but all had recovered pulses before discharge. In Panels (G–I), patients had an absent pulse after the catheterization for a PDA stent (G), attempted ventricular septal defect closure (H) and PDA closure (I) and were started on heparin infusion [Color figure can be viewed at wileyonlinelibrary.com]

validated against ultrasound findings and clinical outcomes, algorithms can be used to develop a smartphone application combining IRT and clinical examination to accurately predict and treat femoral arterial thrombosis.

5 | CONCLUSION

Infrared thermal imaging offers a safe, noncontact and noninvasive adjunctive tool to aid detection and monitoring of femoral arterial thrombosis postcardiac catheterization. Our initial experience shows promise in terms of feasibility, validity, and reliability, however further analysis with larger patient numbers will be necessary to investigate the role and clinical impact of IRT in this context.

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

DATA AVAILABILITY STATEMENT

The data that supports the findings of this study are available in the Supporting Information Material of this article. Additional data are available from the corresponding author upon reasonable request.

ORCID

Sok-Leng Kang http://orcid.org/0000-0003-1556-7395
Lee Benson https://orcid.org/0000-0002-1407-1825

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